

$\frac{1}{6} n \bar{v}$ cross plane A from below
 $\frac{1}{6} n \bar{v}$ cross plane A from above

Molecules experienced their last collision (momentum exchange) a distance λ away from the plane A ($\lambda =$ mean free path).

So an x-momentum balance yields:

$$m_x \uparrow = \frac{1}{6} n \bar{v} [m u_x(y - \lambda)] \quad \text{= momentum transported per unit time per unit area across plane upwards}$$

$$m_x \downarrow = \frac{1}{6} n \bar{v} [m u_x(y + \lambda)]$$

$$\tau = \frac{1}{6} n \bar{v} m [u_x(y - \lambda) - u_x(y + \lambda)] \Rightarrow \text{Taylor series expand}$$

$$\boxed{\tau = \frac{1}{6} n \bar{v} m \left(-2 \frac{\partial u_x}{\partial y} \cdot \lambda \right) = -\mu \frac{\partial u}{\partial y}} \Rightarrow \text{Shear stress on plane A.}$$

Note here, $n \cdot m = \rho$

$$\boxed{\mu = \frac{1}{3} \rho \bar{v} \lambda} \Rightarrow \text{Molecular Viscosity (for a gas)}$$

Note, before we showed that; $\lambda = \frac{k_B T}{\pi d^2 \rho}$ (ideal gas)

$$\bar{v} = \sqrt{\frac{8 k_B T}{\pi m}}$$

Interestingly, back substituting $\bar{v}$ & λ into μ

$$\mu = \frac{1}{3} \rho \sqrt{\frac{8 k_B T}{\pi m}} \cdot \frac{k_B T}{\pi d^2 \rho} \Rightarrow n = \frac{\rho}{k_B T} \Rightarrow \rho = n \cdot m$$

$$\mu = \frac{1}{3} m \frac{\rho}{k_B T} \cdot \sqrt{\frac{8 k_B T}{\pi m}} \cdot \frac{k_B T}{\pi d^2 \rho} = \frac{m}{3 \pi d^2} \sqrt{\frac{8 k_B T}{\pi m}}$$

$$\boxed{\mu \sim T^{1/2}} \neq f(p, \rho)!$$

Interesting since μ increases at higher temperatures for gas. Opposite for liquids.

So for our turbulent viscosity, we can assume a similar form as our molecular viscosity.

$$\boxed{\text{Molecules Diffusing}} \xrightleftharpoons{\text{Analogy}} \boxed{\text{Eddies Diffusing}}$$

$$\mu = \frac{1}{3} \rho \bar{v} \lambda$$

$$\boxed{\mu_t \sim \rho u_* l}$$

mean free path analogous to eddy size
 $\Rightarrow u_* \equiv \text{friction velocity}$
 $l \equiv \text{mixing length or Eddy size.}$

This approximation is known as Prandtl's Mixing Length model. Simplest approach.

Doing a wall coordinate nondimensionalization:

$$u^+ = \frac{\bar{u}}{u_*}, \quad v^+ = \frac{\bar{v}}{u_*}$$

$$x^+ = \frac{x u_*}{\nu}, \quad y^+ = \frac{y u_*}{\nu}$$

We can re-cast $\tau_0 = \rho \frac{\partial \bar{u}}{\partial y} (\nu + \epsilon)$ as: $(u_* = (\frac{\tau_0}{\rho})^{1/2})$

$$\frac{\tau_0}{\rho} = \frac{\partial \bar{u}}{\partial y} (\nu + \epsilon) \Rightarrow \partial \bar{u} = u_* \partial u^+; \quad \partial y = \frac{\partial y^+ \nu}{u_*}$$

$$\frac{\tau_0}{\rho u_*^2} = \frac{\partial u^+}{\partial y^+} \cdot \frac{u_*^2}{\nu} (\nu + \epsilon)$$

$$\boxed{\left(1 + \frac{\epsilon}{\nu}\right) \frac{\partial u^+}{\partial y^+} = 1} \Rightarrow u^+(y^+) \equiv \text{velocity distribution near the wall.}$$

But we still need to know ϵ to solve. Looking back at Prandtl's mixing length model.

$$u_{\tau} = \phi U_* l = \phi \varepsilon \Rightarrow \varepsilon = l \left(\frac{\tau_0}{\rho} \right)^{1/2}$$

$\swarrow$
 $\searrow$ need to determine

Prandtl assumed that the size of turbulent eddies cannot be bigger than their distance from the wall, hence:

$$\boxed{l = Ky} \equiv \text{mixing length}$$

Back substituting

$$\left(1 + \frac{\varepsilon}{U}\right) \frac{\partial u^+}{\partial y^+} = 1$$

In the viscous sublayer, $U \gg \varepsilon$, $\frac{\varepsilon}{U} \ll 1$

$$\int \frac{\partial u^+}{\partial y^+} = \int 1$$

$$\boxed{u^+ = y^+} \equiv \text{Viscous sublayer velocity profile}$$

In the turbulent layer, $\varepsilon \gg U$, $\frac{\varepsilon}{U} \gg 1$

$$\frac{\varepsilon}{U} \frac{\partial u^+}{\partial y^+} = 1 \Rightarrow \text{use mixing length model}$$

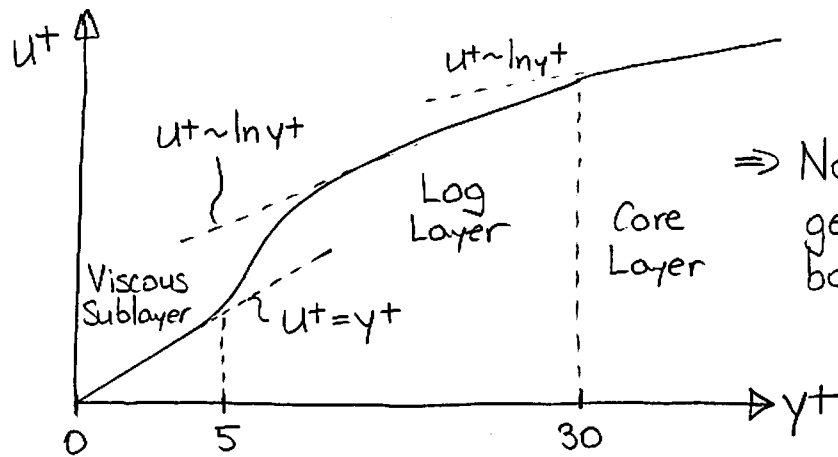
$$\frac{l U_*}{U} \frac{\partial u^+}{\partial y^+} = 1 \Rightarrow l = Ky = K \frac{y^+ U}{U_*}$$

$$Ky^+ \frac{U}{U_*} \cdot \frac{U_*}{U} \frac{\partial u^+}{\partial y^+} = 1$$

$$Ky^+ \frac{\partial u^+}{\partial y^+} = 1 \Rightarrow \int \partial u^+ = \int \frac{dy^+}{y^+ K}$$

$$\boxed{u^+ = \frac{1}{K} \ln y^+ + C} \equiv \text{Log-layer velocity profile.}$$

Note, this turbulent b.l. behaviour is also applicable to internal flows.



⇒ Note, applicable to general turbulent boundary layer flow.

Experimentally verified solutions are:

$$\left. \begin{array}{l} u^+ = y^+ ; \\ u^+ = 2.5 \ln y^+ + 5.5 ; \end{array} \right\} \begin{array}{l} 0 < y^+ < 11.6 \\ y^+ > 11.6 \end{array} \quad \text{Prandtl & Taylor}$$

$$\left. \begin{array}{l} u^+ = y^+ \\ u^+ = 5 \ln y^+ - 3.05 \\ u^+ = 2.5 \ln y^+ + 5.5 \end{array} \right\} \begin{array}{l} 0 < y^+ < 5 \\ 5 < y^+ < 30 \\ y^+ > 30 \end{array} \quad \left. \begin{array}{l} \text{von Karman (1939)} \\ \text{Result plotted above} \end{array} \right\}$$

So to solve for shear stress in turbulent flows, we need to solve for τ_0 using our profiles above. Note:

$$\tau_0 \sim \frac{2\bar{u}}{\partial y} \sim \frac{U_\infty}{\delta} \Rightarrow \text{We need } \delta \text{ for a turbulent b.l.}$$

We can approximate our piecewise profiles with one good solution: $u^+ = f(y^+)$

$$\boxed{u^+ = 8.75(y^+)^{1/7}} \Rightarrow \text{Prandtl's } \neq \text{ power law}$$

Dimensionalizing our equation ($u^+ = f(y^+)$) for $\bar{u} = U_\infty$ at $y = \delta$

$$\frac{\bar{u}}{U_*} = f\left(\frac{y U_*}{\nu}\right) \Rightarrow U_* = \left(\frac{\tau_0}{\rho}\right)^{1/2}$$

$$\text{For } \bar{u} = U_\infty, y = \delta \Rightarrow \boxed{\frac{U_\infty}{(\tau_0/\rho)^{1/2}} = f\left(\frac{\delta}{\nu} \left(\frac{\tau_0}{\rho}\right)^{1/2}\right)}$$

Using Prandtl's $1/7$ power law as f , i.e. $f = 8.75 (y^+)^{1/7}$
 And applying our momentum integral equation:

$$\frac{d}{dx} \int_0^\infty \bar{u} (U_\infty - \bar{u}) dy = \frac{\tau_0}{\rho} \quad (1)$$

$$\frac{U_\infty}{(\tau_0/\rho)^{1/2}} = 8.75 \left(\frac{\delta}{U} \left(\frac{\tau_0}{\rho} \right)^{1/2} \right)^{1/7} \quad (2) \Rightarrow \text{Solve for } \tau_0$$

$$\tau_0 = 0.0225 \rho U_\infty^2 \left(\frac{\delta U_\infty}{U} \right)^{-1/4} \quad (3)$$

Substituting (3) into (1) & integrate to $y = \delta \Rightarrow$ we can now solve.
 Complex method since $\bar{u} = f(\tau_0)$ as well,

$$\boxed{\frac{\delta}{x} = 0.37 \left(\frac{U_\infty x}{U} \right)^{-1/5} = \frac{0.37}{Re_x^{0.2}}} \quad (4) \Rightarrow \text{Turbulent b.l. thickness.}$$

Combining (4) and (2) yields:

$\frac{\tau_0}{\rho U_\infty^2} = \frac{1}{2} C_{f,x} = 0.0296 \left(\frac{U_\infty x}{U} \right)^{-1/5}$	$\Rightarrow$ Local turbulent skin friction coeff. Flat plate.
$\frac{\bar{\tau}_0}{\rho U_\infty^2} = \frac{1}{2} \bar{C}_f = 0.037 \left(\frac{U_\infty x}{U} \right)^{-1/5}$	$\Rightarrow$ Average turbulent skin friction coeff. $10^5 < Re_x < 10^8$

Experiments show that:

$$\boxed{C_{f,x} = 0.37 \left[\log_{10} \left(\frac{U_\infty x}{U} \right) \right]^{-2.584}} \quad 10^5 < Re_x < 10^{10}$$

Note, discrepancy arises between experiment & theory at $Re_x > 10^8$ due to our assumption of $u^+ = f(y^+) \Rightarrow$ Prandtl's $1/7$ Power law.

Turbulent Heat Transfer

If we follow the same procedure as before of time averaging our energy equation:

$$\boxed{\bar{u} \frac{\partial \bar{T}}{\partial x} + \bar{v} \frac{\partial \bar{T}}{\partial y} = \alpha \nabla^2 \bar{T} - \frac{\partial}{\partial x} (\overline{u'T'}) - \frac{\partial}{\partial y} (\overline{v'T'})}$$

↳ Time averaged energy equation for turbulent flow.

For boundary layers: ($u' \sim v'$)

$$\frac{\frac{\partial}{\partial x} (\overline{u'T'})}{\frac{\partial}{\partial y} (\overline{v'T'})} \sim \frac{\frac{\overline{u'T'}}{L}}{\frac{\overline{v'T'}}{\delta_T}} \sim \frac{\delta_T}{L} \ll 1 \Rightarrow \text{so we can drop the } x \text{ term}$$

Our b.l. energy equation becomes:

$$\begin{aligned} \bar{u} \frac{\partial \bar{T}}{\partial x} + \bar{v} \frac{\partial \bar{T}}{\partial y} &= \alpha \underbrace{\frac{\partial^2 \bar{T}}{\partial y^2}}_{= \frac{1}{\rho c_p} \frac{\partial \bar{T}}{\partial y} \left(k \frac{\partial \bar{T}}{\partial y} - \rho c_p \overline{v'T'} \right)} \\ &= \frac{1}{\rho c_p} \frac{\partial \bar{T}}{\partial y} \left(k \frac{\partial \bar{T}}{\partial y} - \rho c_p \overline{v'T'} \right) \end{aligned}$$

Looks analogous to our shear solution a few pages ago.

Using the same reasoning as before, we can show:

$$\boxed{-\rho c_p \overline{v'T'}} = \rho c_p \epsilon_T \frac{\partial \bar{T}}{\partial y} \equiv \text{Eddy Heat Flux}$$

$\epsilon_T \equiv$ eddy thermal diffusivity

$$\boxed{-q''_{\text{app}} = k \frac{\partial \bar{T}}{\partial y} - \rho c_p \overline{v'T'} = \rho c_p (\alpha + \epsilon_T) \frac{\partial \bar{T}}{\partial y}} \equiv \text{Apparent Heat Flux}$$

We can also define:

$$\boxed{Pr_t = \frac{\epsilon}{\epsilon_t}} = \text{turbulent Prandtl \#}$$

The nice thing is we don't have to solve for the heat flux. We can use the Turbulent Colburn Analogy.

$$\boxed{St_x \cdot Pr^{2/3} = \frac{C_{f,x}}{2}} \Rightarrow \text{Colburn Analogy (Did this in Laminar flow)}$$

$$St_x = \frac{Nu_x}{Re_x Pr} = \frac{\bar{h} x / k}{\cancel{\rho U_{\infty} x} \cdot \cancel{(\rho) \frac{c_p}{k}} \cdot \frac{c_p}{k}} = \frac{\bar{h}}{\rho c_p U_{\infty}} = \frac{\bar{q}'' / \Delta T}{\rho c_p U_{\infty}}$$

$$\boxed{Nu_x = \frac{C_{f,x}}{2} Re_x Pr^{1/3}} \Rightarrow \text{We already solved for } C_{f,x}$$

$$C_{f,x} = 0.0592 Re_x^{-1/5} \Rightarrow \text{External turbulent flow over flat plate}$$

Back substituting, we obtain:

$$Nu_x = \frac{0.0592}{2} Re_x^{-1/5} \cdot Re_x \cdot Pr^{1/3}$$

$$\boxed{Nu_x = 0.029 Re_x^{4/5} Pr^{1/3}} \Rightarrow \text{Colburn correlation, } Pr \geq 0.5, Re_x > 5 \times 10^5$$

For the best correlation (most accurate):

$$\boxed{Nu_x = \frac{\left(\frac{C_{f,x}}{2}\right) \cdot Re_x \cdot Pr}{1 + 12.7 \left(\frac{C_{f,x}}{x}\right)^{1/2} (Pr^{2/3} - 1)}} \Rightarrow \text{White correlation}$$

$0.5 \leq Pr \leq 2000$
 $5 \times 10^5 \leq Re_x \leq 10^7$

Using the Colburn result:

$$\boxed{\overline{Nu}_L = \frac{\bar{h} L}{k} = 0.037 Re_L^{4/5} Pr^{1/3}}, Pr \geq 0.5, Re_x > 5 \times 10^5$$

Note, our correlations are valid for both $T_o = \text{constant}$ and $q''|_o = \text{constant}$

For $q''|_o = \text{constant}$, only a 4% difference than the results above,
 Note also $Nu_x = \frac{q''|_o x}{k [T_o(x) - T_{\infty}]} \equiv \text{constant heat flux Nusselt \#}$

Side Note: Proof of the Colburn Analogy:

$$St = \frac{h}{\rho C_p U_\infty}, \quad \alpha = \frac{k}{\rho C_p}$$

$$\left. \begin{aligned} \tau_o &= \mu \frac{\partial \bar{u}}{\partial y} \\ q''_{o, \text{ conv}} &= k \frac{\partial T}{\partial y} \end{aligned} \right\} \frac{\tau_o}{q''_{o, \text{ conv}}} = \frac{\mu}{k} \frac{\partial \bar{u}}{\partial T} \sim \frac{\mu}{k} \frac{U_\infty}{\Delta T}$$

Here, we know $k = \rho C_p \alpha$, $q''_{o, \text{ conv}} = h \Delta T$ and $\tau_o = \frac{1}{2} \rho U_\infty^2 C_{f,x}$
Back substituting:

$$\frac{\tau_o}{q''_{o, \text{ conv}}} = \frac{\frac{1}{2} \rho U_\infty^2 C_{f,x}}{h \Delta T} \sim \frac{\mu U_\infty}{k \Delta T}$$

$$\frac{\frac{1}{2} U_\infty C_{f,x}}{h} \sim \frac{\mu}{\rho} \frac{1}{\rho C_p \alpha} \sim \underbrace{\frac{\mu}{\rho}}_{\nu} \cdot \underbrace{\frac{1}{\rho C_p}}_{Pr}$$

$$\frac{\frac{1}{2} C_{f,x}}{h} \sim Pr \frac{1}{\rho C_p U_\infty} \Rightarrow \frac{1}{2} C_{f,x} \sim \underbrace{\frac{h}{\rho C_p U_\infty}}_{St} \cdot Pr$$

$$\therefore \boxed{St = \frac{h}{\rho C_p U_\infty} \sim \frac{1}{2} \cdot \frac{C_{f,x}}{Pr}} \Rightarrow \text{QED.}$$